

Dose calculation project

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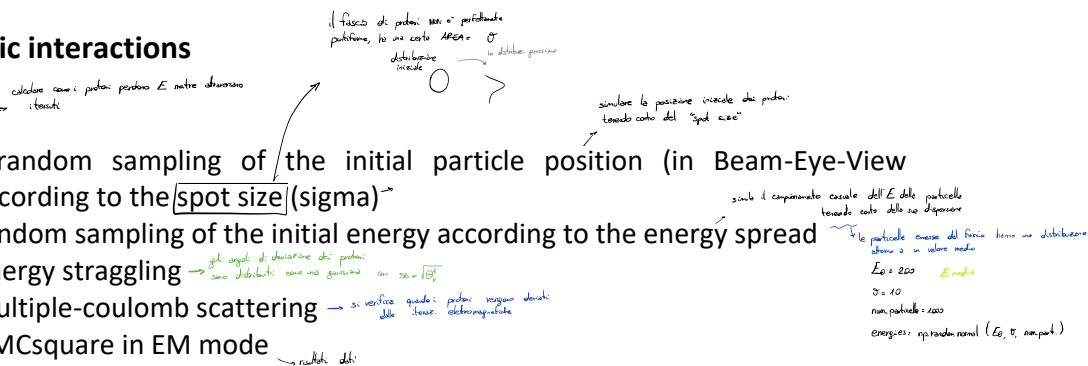
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I. Challenge

The goal is to implement the Monte Carlo simulation by following the steps here-below. All formulas can be found in the paper of Fippel (provided on Teams). You can use MCsquare simulation results as the reference to validate your code.

I.A. Electromagnetic interactions

1. Simulate CSDA
2. Simulate the random sampling of the initial particle position (in Beam-Eye-View coordinates) according to the **spot size** (σ)
3. Simulate the random sampling of the initial energy according to the energy spread
4. Simulate the energy straggling
5. Simulate the multiple-coulomb scattering
6. Compare with MCsquare in EM mode



I.B. Nuclear reactions

1. Compute the macroscopic cross-sections for proton-proton elastic collision, proton-oxygen elastic collision, proton-oxygen inelastic collision.
2. From the macroscopic cross-sections, use basic Monte Carlo sampling techniques to determine at every step the chance of having one of the three aforementioned interactions.
3. Simulate the elastic and inelastic nuclear interactions.
4. Compute the integral depth-dose curve. Compare your results with MCsquare in full simulations mode.

I.C. Testing

1. Set the density for the half-left ($X > 0$) of the phantom to 2 g/cm^3 .
2. Use a realistic spot size (e.g. $\sigma = [0.3\ 0.3]\text{ cm}$).
3. Compute now the integral depth-dose curve and explain the results.

II. One step further (optional)

II.A. Implementation of additional features

1. Implement ICRU compact bone in the code.
2. Simulate in a homogeneous phantom.
3. Simulate in a heterogeneous phantom where voxels along depth with an even index are water and with an odd index are bone.
4. Convert to dose to water “on-the-fly”. Convert also using an energy independent conversion on the obtained dose distributions. What do you observe? Explain.

II.B. Introduce approximations

1. Implement the beam central-axis approximation, with and without sampling of the incident spot size.
2. Test both approaches using the same phantom configuration as section I.C (point 2).
3. Interpret the results.